Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	9 March 2026
Team ID	SWUID202641986
Project Name	Automobile Insurance Fraud Detection System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIN
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Claim Analysis Dashboard	Input claim attributes (age, incident severity, claim amount) via a web form.
FR-4	Fraud Prediction Engine	Display binary prediction result (Fraud/Genuine) using the Random Forest model.
FR-5	Data Visualization	Display charts for incident severity distribution and correlation heatmaps for auditors.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Flask web interface must be intuitive, allowing investigators to enter data without special technical training.
NFR-2	**Security**	Customer data must be encrypted, and only authorized auditors should access prediction results.
NFR-3	**Reliability**	The system must consistently use the saved `.pkl` model to provide stable predictions across different sessions.

NFR-4	**Performance**	The model should return a "Fraud" or "Genuine" prediction within 2 seconds of the user submitting the form.
NFR-5	**Availability**	The web application should be accessible for use by insurance teams 24/7.
NFR-6	**Scalability**	The system should be able to handle an increasing volume of claims as the company's customer base grows.